1. Spread Operator (. . .)

The Spread Operator is used to **expand** elements of an array or properties of an object.

Array Example

```
const arr1 = [10, 20, 30];
const arr2 = [...arr1];

console.log(arr2);
```

Output

```
[10, 20, 30]
```

Merge Arrays

```
const arr1 = [1, 2, 3];
const arr2 = [4, 5, 6];

const result = [...arr1, ...arr2];

console.log(result);
```

Output

```
[1, 2, 3, 4, 5, 6]
```

Add New Elements

```
const numbers = [1, 2, 3];

const updated = [...numbers, 4, 5];

console.log(updated);
```

Output

```
[1, 2, 3, 4, 5]
```

Object Example

```
const user = {
  name: "Dhruv",
  age: 22
};

const newUser = {
  ...user,
  city: "Ahmedabad"
};

console.log(newUser);
```

Output

```
{
  name: "Dhruv",
  age: 22,
  city: "Ahmedabad"
}
```

Update Object Property

```
const user = {
  name: "Dhruv",
  age: 22
};

const updatedUser = {
  ...user,
  age: 23
};

console.log(updatedUser);
```

Output

```
{
  name: "Dhruv",
  age: 23
}
```

2. Rest Operator (. . .)

The Rest Operator is used to **collect multiple values into one variable**.

Function Example

```
function add(...numbers) {
  console.log(numbers);
}

add(10, 20, 30, 40);
```

Output

```
[10, 20, 30, 40]
```

Sum Example

```
function sum(...numbers) {
  let total = 0;

  for (let num of numbers) {
    total += num;
  }

  return total;
}

console.log(sum(10, 20, 30));
```

Output

```
60
```

Array Destructuring with Rest

```
const numbers = [10, 20, 30, 40, 50];

const [first, second, ...remaining] = numbers;

console.log(first);
console.log(second);
console.log(remaining);
```

Output

```
10
20
[30, 40, 50]
```

Object Destructuring with Rest

```
const user = {
  name: "Dhruv",
  age: 22,
  city: "Ahmedabad"
};

const { name, ...others } = user;

console.log(name);
console.log(others);
```

Output

```
Dhruv

{
  age: 22,
  city: "Ahmedabad"
}
```

Difference Between Rest and Spread

Spread Operator	Rest Operator
Expands values	Collects values
Used while copying or merging	Used while receiving values
Converts one array/object into individual values	Converts multiple values into a single array/object
`[...arr]`	`(...args)`

Operators Used with Arrays & Objects in React

1. Spread Operator (. . .)

```
const newArray = [...oldArray];
const newObject = {...oldObject};
```

Used for:

- Copying
- Merging
- Updating State

2. Destructuring Operator

Array Destructuring

```
const colors = ["Red", "Blue", "Green"];

const [first, second] = colors;

console.log(first);
```

Output:

```
Red
```

Object Destructuring

```
const user = {
  name: "Dhruv",
  age: 22
};

const { name, age } = user;

console.log(name);
```

Output

```
Dhruv
```

3. Optional Chaining (`?.`)

Prevents errors when a property doesn't exist.

```
const user = {};

console.log(user?.name);
```

Output

```
undefined
```

4. Nullish Coalescing (`??`)

Provides a default value when value is `null` or `undefined`.

```
const username = null;

console.log(username ?? "Guest");
```

Output

```
Guest
```

5. Array Methods Frequently Used in React

map()

```
const users = ["Dhruv", "Rahul", "Amit"];

users.map((user) => {
  console.log(user);
});
```

filter()

```
const numbers = [10, 20, 30, 40];

const result = numbers.filter((num) => num > 20);

console.log(result);
```

Output:

```
[30, 40]
```

find()

```
const users = [
  { id: 1, name: "Dhruv" },
  { id: 2, name: "Rahul" }
];

const user = users.find((u) => u.id === 2);

console.log(user);
```